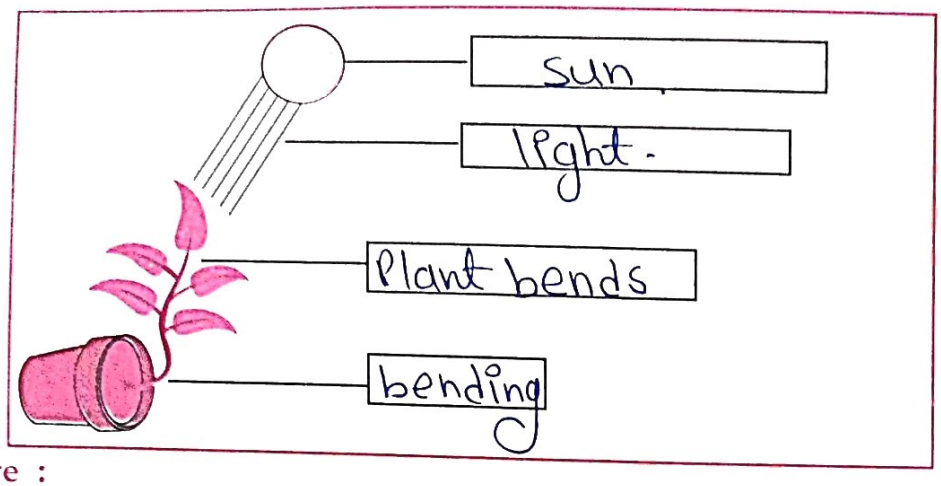
Experiment No.10

Title : To verify that the plants grow in the direction of sunlight.

Material : Small potted plant.

Figure :



Procedure :

1. Take a small potted plant.
2. Keep the potted plant near the window of dark room as shown in figure.
3. keep observing the growth of plant daily for 10 to 15 days .
4. Note down the observations.

Observations :

What and which type of changes did you observe while growing the potted plant ?

The main change in ~~a~~ potted is the speed of growing of plants is very loa.

Inference :

The shoot positive photospism the shoot system of a plant shows positive responsible to the stimulus of light when light focal tip of the shoot speed and given stimulul for the growth of the shoot.

The shoot system of a plant responds to the stimulus of light. When light falls on the part of the plant, a hormone called 'auxin' which is synthesised at the tip of the shoot spreads and gives stimulus to grow that part. Therefore the plants grow in the direction of sunlight.

Multiple Choice Questtions

1. In the shoot system of plants .....hormones are found.
   A. Abscissic Acid B. Gibberellins C. Auxin D. Cytokinins
2. The growth of the plant in the direction of sunlight is .....type of movement.
   A. Hydrotrphic B. Gravitrophic C. Chemotrophic D. Phototrophic
3. The leaflets of minosa plants close on touch. This is ..... type of movement.
   A. Gravitrophic B. Seismotrophic C. Hydrotrophic D. Chemotrophic
4. The movements of sprouts towards gravity shows .....type of movement.
   A. Chemotrophic B. Gravitrophic and Hydrotrophic
   C. Phototrophic D. Electro chemical
5. In fruits and seeds .....hormone is commonly observed.
   A. Gibberellin B. Abscissic acid C. cytokinins D. Auxin
6. Wilting of leaves occurs due to .....
   A. Auxins B. Gibberellins C. Abscissic acid D. Cytokinins

Questions

1. Which different types of movements observed in plants ?

There are two types - Tropism & nustism tropism movements are of four types - phototropism, hydrotopism geotropism and thing metrapism

2. Read the following passage and complete the given table.

The movement or growth of parts of plant in response to the stimulus is called as 'tropism' or ' tropic movement' shoot system of any plant responds to the stimulus of light i.e it grows in the direction of source of light. The root system of plant responds to the stimulus of light and water. Specific parts of plant responds to the stimulus of chemicals released.

Movement	Which parts of plant shows this movement?	Examples
Phototrophic Movement	shoot of Plant	Growth toward the source of light.
Gravitrophic Movement	root of Plant	the minerals from soil.
Hydrotrophic Movement	root of Plant	Attract with specifice type of chemical.
Chemotrophic Movement	Part of Plant	
Electro-chemical Movement		

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